

Development of a Modular Educational and Calculation Software (Calc Uni) for Engineering and Basic Sciences Students

Project Coordinator: Muhammet Can Yolci

Title of the Research Proposal: Development of a Modular Educational and Calculation Software (Calc Uni) for Engineering and Basic Sciences Students

Project Partners:

1.Ahmet Serhat Paçacı (First year Computer Engineering)

2.Esmagül İpek (First year Aeronautical Engineering)

3.Samet Sarıkaya (First year Aeronautical Engineering)

Advisor's Full Name: İlyas Kandemir

Host Institution/Organization: Gebze Technical University (GTU)

Aim and Importance:

First-year students studying in the engineering and basic sciences faculties of universities encounter difficulties in applying and verifying complex formulas in Physics (Mechanics, Electricity and Magnetism), Calculus, and introductory aviation courses. While professional mathematical analysis software available in the market has a steep learning curve for beginners, simple calculators are inadequate for visualizing these complex problems and flawlessly performing the multiple unit conversions that are critical in engineering.

The aim of this project is to develop a desktop-based educational software (Calc Uni) that allows students to quickly solve physics, mathematics, and basic aviation problems, perform reliable conversions between engineering units, verify their results, and grasp the underlying logic of the formulas. The project is of great importance as it minimizes students' calculation errors, enabling them to focus on the theoretical essence of the subject, while also providing a domestic educational technology tool.

Project Methodology and Work Plan:

The project will be carried out in four main phases in accordance with the Agile software development methodology:

- **Phase 1: Requirement Analysis and Algorithm Design** The relevant course topics will be formulated, and the necessary algorithms (flowcharts) will be created to solve these formulas in a computational environment.

- **Phase 2: Software Development (Back-end and Front-end)** The core architecture of the application will be established using the C# language and the .NET framework. The work, energy, harmonic motion, calculus, unit conversion, and aviation panels will be coded in accordance with ObjectOriented Programming (OOP) principles.
- **Phase 3: User Interface Design** A user-friendly and modular interface (GUI) will be designed and integrated with the back-end code to ensure ease of use for the target student group.
- **Phase 4: Testing and Validation (Debugging)** The developed software will be tested using standard problem sets. Debugging processes will be completed by cross-referencing the software outputs with manual calculations to ensure 100% accuracy.

WORK PACKAGES AND PROJECT TIMELINE

	Work Package Name and Objectives	o Be Performed By	Time Frame(... Month)	Success Criteria and Contribution to Project Success
1	Requirement Analysis and Algorithm Design	Muhammet Can Yolci (Project Coordinator)	Month 1	Comprehensive determination of the module flowcharts and approval of the UI prototype. This establishes the foundational architecture of the project.
2	Coding of All Software Modules (Back-end)	Muhammet Can Yolci Ahmet Serhat Paçacı (Team Member)	Month 2	All developed functions yielding 100% identical results to reference tables and manual calculations during console testing. This completes the construction of the application's theoretical infrastructure.
3	Desktop User Interface (GUI) Design and Integration	Muhammet Can Yolci Samet Sarıkaya (Team Member)	Month 3	Seamless transitions between modules and ensuring that data entered the interface triggers the correct functions. This makes the software ready for the end-user experience.
4	System Testing and Debugging	Esmagül İpek (Team Member)	Month 4	Stable operation of the software without crashing against unexpected inputs. This proves the reliability and usability of the application.

RESEARCH FACILITIES TABLE

	Type and Model of Infrastructure/Equipment Available at the Institution	Purpose of Use in the Project
1	Visual Studio IDE and .NET Platform	To be used for developing the software, performing debugging operations, and packaging the application (creating setup).
2	University Academic Databases and Digital Library	To be used for verifying the literature accuracy of aviation standards (ISA tables, etc.) and engineering formulas.

Programming Language Selection Matrix

Evaluation Criteria	Weight	C	C++	Python	C#
Desktop Interface (GUI) Development Ease	3	1	3	2	5
Object-Oriented Programming (OOP) Infrastructure	2	1	5	4	5
Development and Debugging Speed	2	2	3	5	4
Computational Performance (Execution Speed)	2	5	5	2	4
Mathematical Function and Library Support	1	2	4	5	4
WEIGHTED TOTAL SCORE	-	21	39	33	45

RISK MANAGEMENT TABLE

	Major Risks	Risk Management (Plan B)
1	Technical errors (bugs) or library incompatibilities encountered during the software development process.	Isolating errors using a modular programming (OOP) structure, conducting regular version control (GitHub/Git), and maintaining the ability to revert to previous stable versions.
2	Instability of the software on different hardware or unexpected user errors during the testing phase.	Increasing error-handling (Try-Catch) blocks, optimizing the minimum system requirements of the software, and expanding the test group to extend the feedback period.

EXPECTED WIDESPREAD IMPACT OF THE RESEARCH PROPOSAL TABLE

	Types of Widespread Impact	Expected Outputs, Results, and Impacts from the Proposed Research
1	Scientific/Academic	Course Material Support Tool: It is aimed to present the developed software for academic use as a digital supplementary tool in first-year basic physics, calculus, and introductory aviation courses, allowing students to verify their manual solutions and test the logic behind the formulas.
2	Economic/Commercial/Social	Product Output and Social Benefit: An easy-to-use, domestic desktop calculation tool (Calc Uni) will be produced for engineering students as an alternative to complex commercial software (e.g., MATLAB). In this way, basic calculation and unit conversion errors, especially for first-year students, will be minimized. Sustainable Infrastructure: Since the software will be coded with Object-Oriented Programming (OOP) and a modular structure, an open-ended database and software framework will be brought to the university, allowing new modules (e.g., a thermodynamics panel) to be easily added according to future needs after the completion of the project.

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